

Лабораторная работа №15

Динамическая маршрутизация

Владимир Базлов

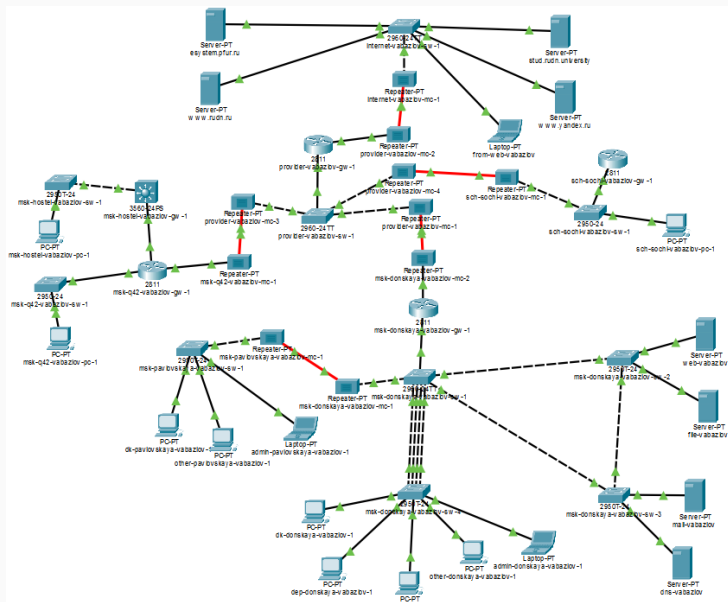
16 мая 2026

Российский университет дружбы народов, Москва, Россия

Цель работы

Настроить динамическую маршрутизацию между территориями организации и проверить изменение маршрута при отказе одного из каналов связи.

Топология сети



Настройка OSPF

```
nsk-donskaya-vabazlov-gw-1>en
Password:
nsk-donskaya-vabazlov-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
nsk-donskaya-vabazlov-gw-1(config)#
nsk-donskaya-vabazlov-gw-1(config)#router ospf 1
nsk-donskaya-vabazlov-gw-1(config-router)#router-id 10.128.254.1
nsk-donskaya-vabazlov-gw-1(config-router)#net 10.0.0.0 0.255.255.255 area 0
nsk-donskaya-vabazlov-gw-1(config-router)#EXIT
nsk-donskaya-vabazlov-gw-1(config)#EXIT
nsk-donskaya-vabazlov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

nsk-donskaya-vabazlov-gw-1#wr mem
Building configuration...
[OK]
nsk-donskaya-vabazlov-gw-1#
```

Рис. 2: Настройка OSPF на msk-donskaya-gw-1

```
msk-q42-vabazlov-gw-1>en
Password:
msk-q42-vabazlov-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-q42-vabazlov-gw-1(config)#
msk-q42-vabazlov-gw-1(config)#router ospf 1
msk-q42-vabazlov-gw-1(config-router)#router-id 10.128.254.2
msk-q42-vabazlov-gw-1(config-router)#net 10.0.0.0 0.255.255.255 area 0
msk-q42-vabazlov-gw-1(config-router)#exit
msk-q42-vabazlov-gw-1(config)#exit
msk-q42-vabazlov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-q42-vabazlov-gw-1#wr mem
Building configuration...
[OK]
msk-q42-vabazlov-gw-1#
01:09:00: %OSPF-5-ADJCHG: Process 1, Nbr 10.128.254.1 on FastEthernet0/1.5 from LOADING to FULL, Loading Done

msk-q42-vabazlov-gw-1#
```

Рис. 3: Настройка OSPF на msk-q42-gw-1


```
msk-hostel-vabazlov-gw-1>en
Password:
msk-hostel-vabazlov-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-hostel-vabazlov-gw-1(config)#
msk-hostel-vabazlov-gw-1(config)#router ospf 1
msk-hostel-vabazlov-gw-1(config-router)#router-id 10.128.254.3
msk-hostel-vabazlov-gw-1(config-router)#network 10.0.0.0 0.255.255.255 area 0
msk-hostel-vabazlov-gw-1(config-router)#exit
msk-hostel-vabazlov-gw-1(config)#exit
msk-hostel-vabazlov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-hostel-vabazlov-gw-1#wr mem
Building configuration...
[OK]
msk-hostel-vabazlov-gw-1#
01:09:41: %OSPF-5-ADJCHG: Process 1, Nbr 10.128.254.2 on Vlan202 from LOADING to FULL, Loading Done
msk-hostel-vabazlov-gw-1#
```

Рис. 4: Настройка OSPF на msk-hostel-gw-1

```
sch-sochi-vabazlov-gw-1>en
Password:
sch-sochi-vabazlov-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
sch-sochi-vabazlov-gw-1(config)#router ospf 1
sch-sochi-vabazlov-gw-1(config-router)#router-id 10.128.254.4
sch-sochi-vabazlov-gw-1(config-router)#net 10.0.0.0 0.255.255.255 area 0
sch-sochi-vabazlov-gw-1(config-router)#exit
sch-sochi-vabazlov-gw-1(config)#exit
sch-sochi-vabazlov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

sch-sochi-vabazlov-gw-1#wr mem
Building configuration...
[OK]
sch-sochi-vabazlov-gw-1#
01:06:55: %OSPF-5-ADJCHG: Process 1, Nbr 10.128.254.1 on FastEthernet0/0.6 from LOADING to FULL, Loading Done

sch-sochi-vabazlov-gw-1#
```

Рис. 5: Настройка OSPF на sch-sochi-gw-1

Прямая связь q42 — Сочи

```
provider-vabazlov-sw-1>en
Password:
provider-vabazlov-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
provider-vabazlov-sw-1(config)#vlan 7
provider-vabazlov-sw-1(config-vlan)#name q42-sochi
provider-vabazlov-sw-1(config-vlan)#int vlan 7
provider-vabazlov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan7, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan7, changed state to up

provider-vabazlov-sw-1(config-if)#no sh
provider-vabazlov-sw-1(config-if)#
provider-vabazlov-sw-1(config-if)#exit
provider-vabazlov-sw-1(config)#exit
provider-vabazlov-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

provider-vabazlov-sw-1#wr mem
Building configuration...
[OK]
provider-vabazlov-sw-1#
```

Рис. 6: Создание VLAN 7 на коммутаторе провайдера

Настройка подинтерфейса q42

```
msk-q42-vabazlov-gw-1#
msk-q42-vabazlov-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-q42-vabazlov-gw-1(config)#int f0/1.7
msk-q42-vabazlov-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/1.7, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.7, changed state to up

msk-q42-vabazlov-gw-1(config-subif)#enc dot1q 7
msk-q42-vabazlov-gw-1(config-subif)#ip addr 10.128.255.9 255.255.255.252
msk-q42-vabazlov-gw-1(config-subif)#descr sochi
msk-q42-vabazlov-gw-1(config-subif)#exit
msk-q42-vabazlov-gw-1(config)#exit
msk-q42-vabazlov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-q42-vabazlov-gw-1#wr mem
Building configuration...
[OK]
msk-q42-vabazlov-gw-1#
```

Рис. 7: Настройка подинтерфейса на msk-q42-gw-1

```
sch-sochi-vabazlov-sw-1>en
Password:
sch-sochi-vabazlov-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
sch-sochi-vabazlov-sw-1(config)#
sch-sochi-vabazlov-sw-1(config)#vlan 7
sch-sochi-vabazlov-sw-1(config-vlan)#name q42-sochi
sch-sochi-vabazlov-sw-1(config-vlan)#int vlan 7
sch-sochi-vabazlov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan7, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan7, changed state to up

sch-sochi-vabazlov-sw-1(config-if)#no sh
sch-sochi-vabazlov-sw-1(config-if)#exit
sch-sochi-vabazlov-sw-1(config)#exit
sch-sochi-vabazlov-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

sch-sochi-vabazlov-sw-1#wr mem
Building configuration...
[OK]
sch-sochi-vabazlov-sw-1#
```

Рис. 8: Создание VLAN 7 на коммутаторе филиала

Настройка подинтерфейса Сочи

```
sch-sochi-vabazlov-gw-1#
sch-sochi-vabazlov-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
sch-sochi-vabazlov-gw-1(config)#
sch-sochi-vabazlov-gw-1(config)#int f0/0.7
sch-sochi-vabazlov-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.7, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.7, changed state to up

sch-sochi-vabazlov-gw-1(config-subif)#enc dot1q 7
sch-sochi-vabazlov-gw-1(config-subif)#ip addr 10.128.255.10 255.255.255.252
sch-sochi-vabazlov-gw-1(config-subif)#descr q
01:10:20: %OSPF-5-ADJCHG: Process 1, Nbr 10.128.254.2 on FastEthernet0/0.7 from LOADING to FULL, Loading Done
42
sch-sochi-vabazlov-gw-1(config-subif)#
sch-sochi-vabazlov-gw-1(config-subif)#exit
sch-sochi-vabazlov-gw-1(config)#exit
sch-sochi-vabazlov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

sch-sochi-vabazlov-gw-1#wr mem
Building configuration...
[OK]
sch-sochi-vabazlov-gw-1#
```

Рис. 9: Настройка подинтерфейса на sch-sochi-gw-1

Проверка маршрутизации

Проверка msk-donskaya-gw-1

```
nsk-donskaya-vabazlov-gw-1#sh ip ospf nei

Neighbor ID      Pri   State           Dead Time   Address        Interface
10.128.254.2      1    FULL/BDR        00:00:31    10.128.255.2   FastEthernet0/1.5
10.128.254.4      1    FULL/BDR        00:00:32    10.128.255.6   FastEthernet0/1.6

nsk-donskaya-vabazlov-gw-1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 198.51.100.1 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 24 subnets, 4 masks
C       10.128.0.0/24 is directly connected, FastEthernet0/0.3
L       10.128.0.1/32 is directly connected, FastEthernet0/0.3
C       10.128.1.0/24 is directly connected, FastEthernet0/0.2
L       10.128.1.1/32 is directly connected, FastEthernet0/0.2
C       10.128.3.0/24 is directly connected, FastEthernet0/0.101
L       10.128.3.1/32 is directly connected, FastEthernet0/0.101
C       10.128.4.0/24 is directly connected, FastEthernet0/0.102
L       10.128.4.1/32 is directly connected, FastEthernet0/0.102
C       10.128.5.0/24 is directly connected, FastEthernet0/0.103
L       10.128.5.1/32 is directly connected, FastEthernet0/0.103
C       10.128.6.0/24 is directly connected, FastEthernet0/0.104
L       10.128.6.1/32 is directly connected, FastEthernet0/0.104
C       10.128.255.0/30 is directly connected, FastEthernet0/1.5
L       10.128.255.1/32 is directly connected, FastEthernet0/1.5
C       10.128.255.4/30 is directly connected, FastEthernet0/1.6
L       10.128.255.5/32 is directly connected, FastEthernet0/1.6
O       10.128.255.8/30 [110/2] via 10.128.255.2, 00:00:35, FastEthernet0/1.5
        [110/2] via 10.128.255.6, 00:00:35, FastEthernet0/1.6
S       10.129.0.0/16 [1/0] via 10.128.255.2
O       10.129.0.0/24 [110/2] via 10.128.255.2, 00:06:25, FastEthernet0/1.5
O       10.129.1.0/24 [110/2] via 10.128.255.2, 00:05:14, FastEthernet0/1.5
O       10.129.128.0/24 [110/3] via 10.128.255.2, 00:05:04, FastEthernet0/1.5
S       10.130.0.0/16 [1/0] via 10.128.255.6
O       10.130.0.0/24 [110/2] via 10.128.255.6, 00:04:05, FastEthernet0/1.6
O       10.130.1.0/24 [110/2] via 10.128.255.6, 00:04:05, FastEthernet0/1.6
    198.51.100.0/24 is variably subnetted, 2 subnets, 2 masks
C       198.51.100.0/28 is directly connected, FastEthernet0/1.4
L       198.51.100.2/32 is directly connected, FastEthernet0/1.4
S*    0.0.0.0/0 [1/0] via 198.51.100.1

nsk-donskaya-vabazlov-gw-1#
```

```
msk-q42-vabazlov-gw-1#sh ip ospf nei
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.128.254.4	1	FULL/BDR	00:00:30	10.128.255.10	FastEthernet0/1.7
10.128.254.1	1	FULL/DR	00:00:36	10.128.255.1	FastEthernet0/1.5
10.128.254.3	1	FULL/BDR	00:00:33	10.129.1.2	FastEthernet1/0.202

```
msk-q42-vabazlov-gw-1#sh ip route
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
 i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
 * - candidate default, U - per-user static route, o - ODR
 P - periodic downloaded static route

Gateway of last resort is 10.128.255.1 to network 0.0.0.0

```

10.0.0.0/8 is variably subnetted, 19 subnets, 4 masks
O    10.128.0.0/24 [110/2] via 10.128.255.1, 00:06:58, FastEthernet0/1.5
O    10.128.1.0/24 [110/2] via 10.128.255.1, 00:06:58, FastEthernet0/1.5
O    10.128.3.0/24 [110/2] via 10.128.255.1, 00:06:58, FastEthernet0/1.5
O    10.128.4.0/24 [110/2] via 10.128.255.1, 00:06:58, FastEthernet0/1.5
O    10.128.5.0/24 [110/2] via 10.128.255.1, 00:06:58, FastEthernet0/1.5
O    10.128.6.0/24 [110/2] via 10.128.255.1, 00:06:58, FastEthernet0/1.5
C    10.128.255.0/30 is directly connected, FastEthernet0/1.5
L    10.128.255.2/32 is directly connected, FastEthernet0/1.5
O    10.128.255.4/30 [110/2] via 10.128.255.1, 00:01:08, FastEthernet0/1.5
      [110/2] via 10.128.255.10, 00:01:08, FastEthernet0/1.7
C    10.128.255.8/30 is directly connected, FastEthernet0/1.7
L    10.128.255.9/32 is directly connected, FastEthernet0/1.7
C    10.129.0.0/24 is directly connected, FastEthernet0/0.201
L    10.129.0.1/32 is directly connected, FastEthernet0/0.201
C    10.129.1.0/24 is directly connected, FastEthernet1/0.202
L    10.129.1.1/32 is directly connected, FastEthernet1/0.202
S    10.129.128.0/17 [1/0] via 10.129.1.2
O    10.129.128.0/24 [110/2] via 10.129.1.2, 00:05:47, FastEthernet1/0.202
O    10.130.0.0/24 [110/2] via 10.128.255.10, 00:01:08, FastEthernet0/1.7
O    10.130.1.0/24 [110/2] via 10.128.255.10, 00:01:08, FastEthernet0/1.7
S*   0.0.0.0/0 [1/0] via 10.128.255.1

```

```
msk-hostel-vabazlov-gw-1#sh ip ospf nei
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.128.254.2	1	FULL/DR	00:00:32	10.129.1.1	Vlan202

```
msk-hostel-vabazlov-gw-1#sh ip route
```

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is 10.129.1.1 to network 0.0.0.0

```
    10.0.0.0/8 is variably subnetted, 14 subnets, 2 masks
O       10.128.0.0/24 [110/3] via 10.129.1.1, 00:06:14, Vlan202
O       10.128.1.0/24 [110/3] via 10.129.1.1, 00:06:14, Vlan202
O       10.128.3.0/24 [110/3] via 10.129.1.1, 00:06:14, Vlan202
O       10.128.4.0/24 [110/3] via 10.129.1.1, 00:06:14, Vlan202
O       10.128.5.0/24 [110/3] via 10.129.1.1, 00:06:14, Vlan202
O       10.128.6.0/24 [110/3] via 10.129.1.1, 00:06:14, Vlan202
O       10.128.255.0/30 [110/2] via 10.129.1.1, 00:06:14, Vlan202
O       10.128.255.4/30 [110/3] via 10.129.1.1, 00:05:01, Vlan202
O       10.128.255.8/30 [110/2] via 10.129.1.1, 00:01:36, Vlan202
O       10.129.0.0/24 [110/2] via 10.129.1.1, 00:06:14, Vlan202
C       10.129.1.0/24 is directly connected, Vlan202
C       10.129.128.0/24 is directly connected, Vlan301
O       10.130.0.0/24 [110/3] via 10.129.1.1, 00:01:36, Vlan202
O       10.130.1.0/24 [110/3] via 10.129.1.1, 00:01:36, Vlan202
S*     0.0.0.0/0 [1/0] via 10.129.1.1
```

```
msk-hostel-vabazlov-gw-1#
```

Проверка sch-sochi-gw-1

```
sch-socni-vabazlov-gw-1#
sch-sochi-vabazlov-gw-1#sh ip ospf nei

Neighbor ID      Pri   State           Dead Time   Address        Interface
10.128.254.2      1    FULL/DR         00:00:30    10.128.255.9   FastEthernet0/0.7
10.128.254.1      1    FULL/DR         00:00:36    10.128.255.5   FastEthernet0/0.6
sch-sochi-vabazlov-gw-1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.128.255.5 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 18 subnets, 3 masks
O       10.128.0.0/24 [110/2] via 10.128.255.5, 00:05:29, FastEthernet0/0.6
O       10.128.1.0/24 [110/2] via 10.128.255.5, 00:05:29, FastEthernet0/0.6
O       10.128.3.0/24 [110/2] via 10.128.255.5, 00:05:29, FastEthernet0/0.6
O       10.128.4.0/24 [110/2] via 10.128.255.5, 00:05:29, FastEthernet0/0.6
O       10.128.5.0/24 [110/2] via 10.128.255.5, 00:05:29, FastEthernet0/0.6
O       10.128.6.0/24 [110/2] via 10.128.255.5, 00:05:29, FastEthernet0/0.6
O       10.128.255.0/30 [110/2] via 10.128.255.5, 00:02:09, FastEthernet0/0.6
           [110/2] via 10.128.255.9, 00:02:09, FastEthernet0/0.7
C       10.128.255.4/30 is directly connected, FastEthernet0/0.6
L       10.128.255.6/32 is directly connected, FastEthernet0/0.6
C       10.128.255.8/30 is directly connected, FastEthernet0/0.7
L       10.128.255.10/32 is directly connected, FastEthernet0/0.7
O       10.129.0.0/24 [110/2] via 10.128.255.9, 00:02:09, FastEthernet0/0.7
O       10.129.1.0/24 [110/2] via 10.128.255.9, 00:02:09, FastEthernet0/0.7
O       10.129.128.0/24 [110/3] via 10.128.255.9, 00:02:09, FastEthernet0/0.7
C       10.130.0.0/24 is directly connected, FastEthernet0/0.401
L       10.130.0.1/32 is directly connected, FastEthernet0/0.401
C       10.130.1.0/24 is directly connected, FastEthernet0/0.402
L       10.130.1.1/32 is directly connected, FastEthernet0/0.402
S*    0.0.0.0/0 [1/0] via 10.128.255.5
```

```
sch-sochi-vabazlov-gw-1#
```

Проверка ICMP

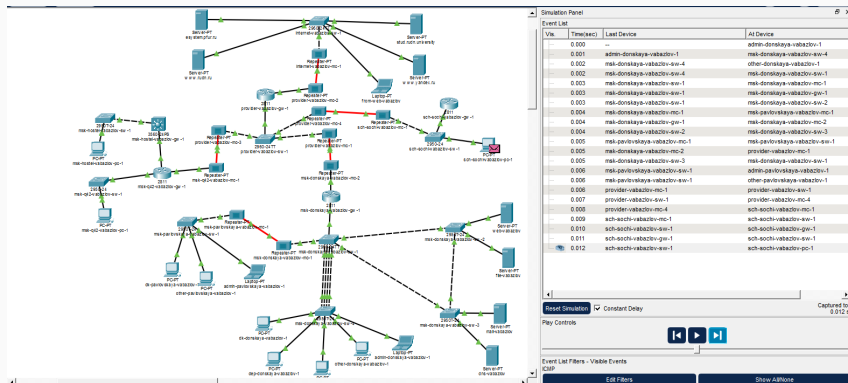


Рис. 14: Прохождение ICMP-пакета до компьютера в Сочи

```
provider-vabazlov-sw-1(config)#  
provider-vabazlov-sw-1(config)#int vlan 6  
provider-vabazlov-sw-1(config-if)#sh  
  
provider-vabazlov-sw-1(config-if)#  
%LINK-5-CHANGED: Interface Vlan6, changed state to administratively down  
  
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan6, changed state to down  
  
provider-vabazlov-sw-1(config-if)#exit  
provider-vabazlov-sw-1(config)#  
provider-vabazlov-sw-1(config)#
```

Рис. 15: Отключение интерфейса Vlan6

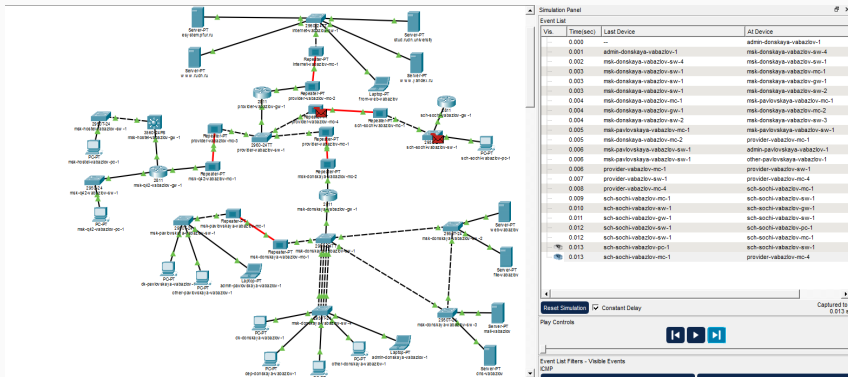


Рис. 16: Изменение маршрута после отключения Vlan6


```
provider-vabazlov-sw-1(config)#
provider-vabazlov-sw-1(config)#vlan 6
provider-vabazlov-sw-1(config-vlan)#name sochi
provider-vabazlov-sw-1(config-vlan)#int vlan 6
provider-vabazlov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan6, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan6, changed state to up

provider-vabazlov-sw-1(config-if)#no sh
provider-vabazlov-sw-1(config-if)#exit
provider-vabazlov-sw-1(config)#exit
provider-vabazlov-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

provider-vabazlov-sw-1#wr mem
Building configuration...
[OK]
provider-vabazlov-sw-1#
```

Рис. 17: Восстановление интерфейса Vlan6

Проверка после восстановления

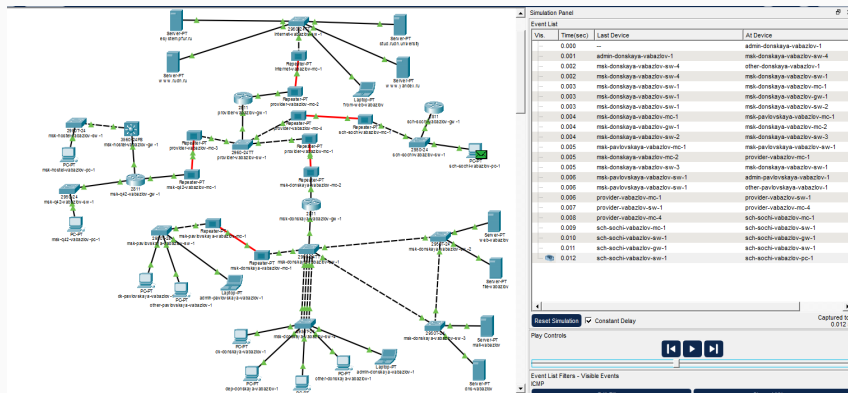


Рис. 18: Прохождение ICMP-пакета после восстановления Vlan6

Итоги работы

В ходе лабораторной работы:

- настроен протокол OSPF на маршрутизаторах организации;
- создана прямая связь между сетью квартала 42 и филиалом в Сочи;
- настроена VLAN 7 и подинтерфейсы с инкапсуляцией 802.1Q;
- проверены OSPF-соседства и таблицы маршрутизации;
- исследовано прохождение ICMP-пакета в режиме Simulation;
- проверено изменение маршрута после отключения VLAN 6;
- восстановлен основной канал связи через VLAN 6.